

AN EMPIRICAL STRESS-INTENSITY FACTOR EQUATION FOR THE SURFACE CRACK

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Abstract—This paper presents an empirical stress-intensity factor equation for a surface crack as a function of parametric angle, crack depth, crack length, plate thickness and plate width for tension and bending loads. The stress-intensity factors used to develop the equation were obtained from a previous three-dimensional, finite-element analysis of semielliptical surface cracks in finite elastic plates subjected to tension or bending loads. A wide range of configuration parameters was included in the equation. The ratios of crack length to plate thickness and the ratios of crack depth to crack length ranged from 0 to 1.0. The effects of plate width on stress-intensity variations along the crack front were also included.

The equation was used to predict patterns of surface-crack growth under tension or bending fatigue loads. The equation was also used to correlate surface-crack fracture data for a brittle epoxy material within ± 10 percent for a wide range of crack shapes and crack sizes.

NOTATION

a	depth of surface crack, mm
b	half-width of cracked plate, mm
C_A, C_B	crack-growth coefficients (see eqns (A1) and (A2))
c	half-length of surface crack, mm
F	stress-intensity boundary-correction factor
h	half-length of cracked plate, mm
K_I	mode I stress-intensity factor, $\text{kN/m}^{3/2}$
K_{IC}	elastic fracture toughness, $\text{kN/m}^{3/2}$
M	applied bending moment, $N \cdot m$
N	number of cycles
n	exponent in equation for crack-growth rate
Q	shape factor for elliptical crack
S_b	remote bending stress on outer fiber, $3M/bt^2$, Pa
S_t	remote uniform-tension stress, Pa
t	plate thickness, mm
ΔK	stress-intensity factor range, $\text{kN/m}^{3/2}$
ϕ	parametric angle of the ellipse, deg.

INTRODUCTION

SURFACE cracks are common flaws in many structural components. Accurate stress analyses of these surface-cracked components are needed for reliable prediction of their crack-growth rates and fracture strengths. However, because of the complexities of such problems, exact solutions are not available. Investigators have used experimental or approximate analytical methods to obtain stress-intensity factors for surface cracks under tension or bending loads. For a semielliptical surface crack in a plate of finite thickness (Fig. 1), Smith and Alavi[1], Smith and Sorensen[2] and Kobayashi *et al.*[3] used the alternating method to obtain the stress-intensity factor variations along the crack front for various crack shapes. Kathiresan[4] and Raju and Newman[5, 6] used the finite-element method to obtain the same information. These results were presented in the form of curves or tables. However, for ease of computation, results in the form of an equation are preferable.

The present paper presents an empirical equation for the stress-intensity factors for a surface crack as a function of parametric angle, crack depth, crack length, plate thickness and plate width for tension and bending loads. The equation was based on the stress-intensity factors obtained from a three-dimensional, finite-element analysis of semielliptical surface cracks in finite elastic plates subjected to tension or bending loads[7]. The present empirical equation covers a wide range of configuration parameters. The ratios of crack depth to plate thickness and the ratios of crack depth to crack length ranged from 0 to 1.0. The effects of plate width on stress-intensity factor variations along the crack front were also included.

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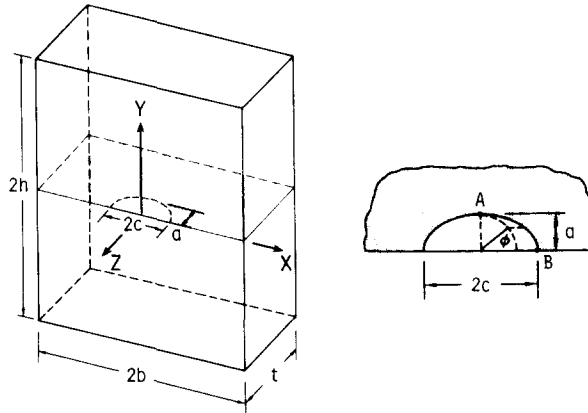


Fig. 1. Surface crack in a finite plate.

In Appendix A, the equation was used to predict surface-crack-growth patterns under tension or bending fatigue loads. These predicted patterns are also compared with measurements reported in the literature for steel, titanium alloy and aluminum alloy materials. In Appendix B, the maximum stress-intensity values from the equation were also used to correlate surface-crack fracture data reported in the literature for a brittle epoxy material.

STRESS-INTENSITY FACTOR EQUATION FOR THE SURFACE CRACK

An empirical equation for the stress-intensity factors for a surface crack in a finite plate subjected to tension and bending loads has been fitted to the finite-element results from Raju and Newman[5-7] for a/c values from 0.2 to 1.0. To account for the limiting behavior as a/c approaches zero, the results of Gross and Srawley[8] for a single-edge crack have also been used. Two types of loads were applied to the surface-cracked plate: remote uniform tension and remote bending. The remote uniform-tension stress is S_t in Fig. 2(a); the remote outer-fiber bending stress S_b in Fig. 2(b) is calculated from the applied bending moment M . The stress-intensity factor equation for combined tension and bending loads is

$$K_I = (S_t + HS_b) \sqrt{\pi \frac{a}{Q}} F\left(\frac{a}{t}, \frac{a}{c}, \frac{c}{b}, \phi\right) \quad (1)$$

for $0 < a/c \leq 1.0$, $0 \leq a/t < 1.0$, $c/b < 0.5$ and $0 \leq \phi \leq \pi$. A useful approximation for Q ,

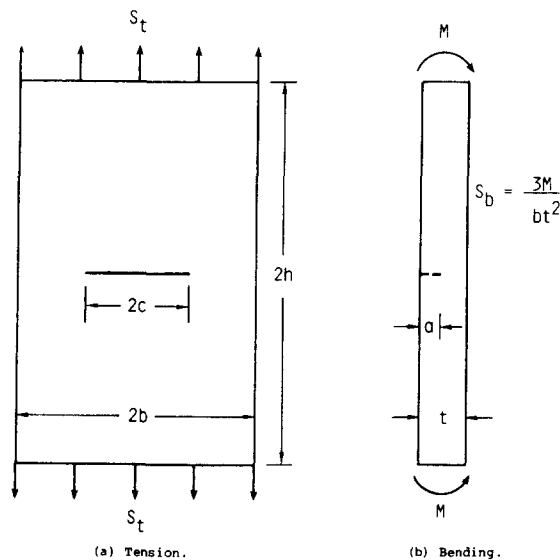


Fig. 2. Surface-cracked plate subjected to tension or bending loads.

developed by Rawe and used in Ref. [9], is

$$Q = 1 + 1.464 \left(\frac{a}{c} \right)^{1.65} \quad \left(\frac{a}{c} \leq 1 \right). \quad (2)$$

The functions F and H are defined so that the boundary-correction factor for tension is equal to F and the boundary-correction factor for bending is equal to the product of H and F . The function F was obtained from a systematic curve-fitting procedure by using double-series polynomials in terms of a/c , a/t , and angular functions of ϕ . The choice of functions was based on engineering judgment. The function F was taken to be

$$F = \left[M_1 + M_2 \left(\frac{a}{t} \right)^2 + M_3 \left(\frac{a}{t} \right)^4 \right] f_\phi g f_w \quad (3)$$

where

$$M_1 = 1.13 - 0.09 \left(\frac{a}{c} \right) \quad (4)$$

$$M_2 = -0.54 + \frac{0.89}{0.2 + (a/c)} \quad (5)$$

$$M_3 = 0.5 - \frac{1.0}{0.65 + (a/c)} + 14 \left(1.0 - \frac{a}{c} \right)^{24} \quad (6)$$

$$g = 1 + \left[0.1 + 0.35 \left(\frac{a}{t} \right)^2 \right] (1 - \sin \phi)^2. \quad (7)$$

The function f_ϕ , an angular function from the embedded elliptical-crack solution[10], is

$$f_\phi = \left[\left(\frac{a}{c} \right)^2 \cos^2 \phi + \sin^2 \phi \right]^{1/4}. \quad (8)$$

The function f_w , a finite-width correction from Ref. [11], is

$$f_w = \left[\sec \left(\frac{\pi c}{2b} \sqrt{\frac{a}{t}} \right) \right]^{1/2}. \quad (9)$$

The function H , developed herein also by curve fitting and engineering judgment, has the form

$$H = H_1 + (H_2 - H_1) \sin^p \phi \quad (10)$$

where

$$p = 0.2 + \frac{a}{c} + 0.6 \frac{a}{t} \quad (11)$$

$$H_1 = 1 - 0.34 \frac{a}{t} - 0.11 \frac{a}{c} \left(\frac{a}{t} \right) \quad (12)$$

$$H_2 = 1 + G_1 \left(\frac{a}{t} \right) + G_2 \left(\frac{a}{t} \right)^2. \quad (13)$$

In this equation for H_2 ,

$$G_1 = -1.22 - 0.12 \frac{a}{c} \quad (14)$$

$$G_2 = 0.55 - 1.05 \left(\frac{a}{c} \right)^{0.75} + 0.47 \left(\frac{a}{c} \right)^{1.5} \quad (15)$$

For all combinations of parameters investigated and $a/t \leq 0.8$, eqn (1) was within $\pm 5\%$ of the finite-element results and the single-edge crack solution. (Herein, "percent error" is defined as the difference between eqn (1) and the finite-element results normalized by the maximum value for that particular case. This definition is necessary, especially for the case of bending, for which the stress-intensity factor ranges from positive to negative along the crack front.) For $a/t > 0.8$, the accuracy of eqn (1) has not been established. However, its use in that range appears to be supported by estimates based on the concept of an equivalent through crack. Results from eqn (1) for tension and bending are shown in Figs. 3 and 4, respectively, with the stress-intensity factor plotted as a function of ϕ for several combinations of a/c and a/t to illustrate the characteristics of the equation. The negative stress-intensity factors shown in Fig. 4 are of course, meaningful only in the presence of sufficient tensile loading to prevent contact between the crack surfaces.

Equation (1) was used in Appendix A to predict the growth patterns of surface cracks under tension and bending fatigue loads. The predicted growth patterns are in good agreement with previously published experimental measurements made on steel, titanium alloy, and aluminum alloy material. The maximum stress-intensity values from eqn (1) were also used in Appendix B to correlate surface-crack fracture data from the literature for a brittle epoxy material under tension loads. In these data, the ratios of crack depth to plate thickness ranged from 0.15 to 1.0 and the ratios of crack depth to crack length ranged from 0.3 to 0.84. The equation correlated 95% of the data analyzed to within $\pm 10\%$ of the calculated failure stress.

CONCLUDING REMARKS

Previously obtained stress-intensity factors from three-dimensional finite-element analyses were used to develop an empirical equation for the stress-intensity factor for both tension and bending loads. The equation applies, for any parametric angle, ratios of crack depth to crack length ranging from 0 to 1.0, ratios of crack depth to plate thickness ranging from 0 to 1.0, and ratios of crack length to plate width less than 0.5. For all configurations for which ratios of crack depth to plate thickness do not exceed 0.8, the equation is within $\pm 5\%$ of the finite-element results and the single-edge crack solution. For ratios greater than 0.8, no solutions are

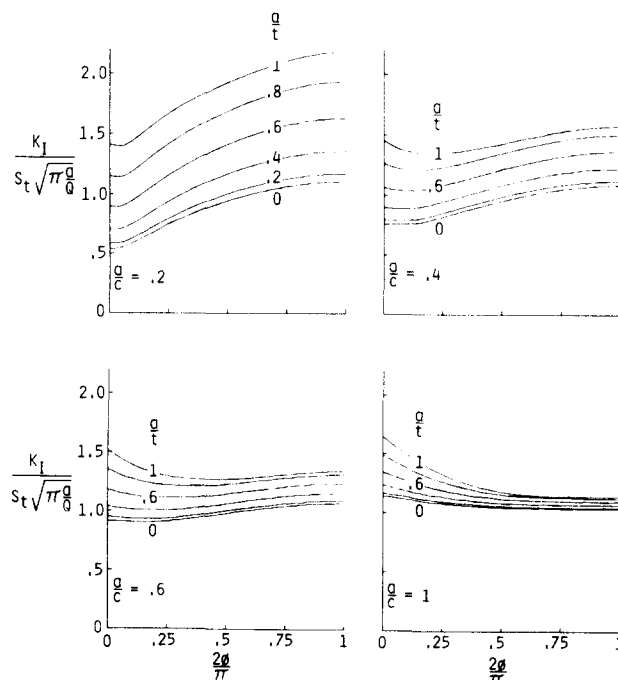


Fig. 3. Typical results from the stress-intensity factor equation (eqn 1) for a semielliptical surface crack in a plate under tension. ($c/b = 0$.)

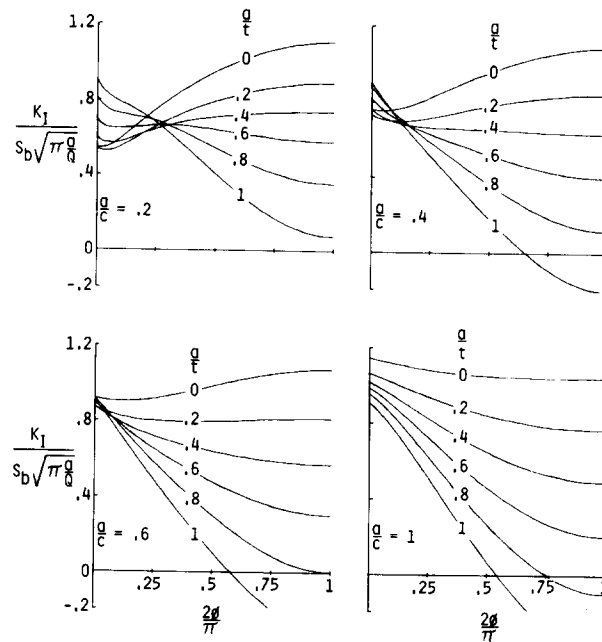


Fig. 4. Typical results from the stress-intensity factor equation (eqn 1) for a semielliptical surface crack in a plate under bending. ($c/b = 0$.)

available for direct comparison; however, the equation appears reasonable on the basis of engineering estimates.

The wide-range equation was used in Appendix A to predict the growth patterns of surface cracks under tension and bending fatigue loads. The predicted growth patterns were in good agreement with previously published experimental measurements made on steel, titanium alloy and aluminum alloy material. The equation was also used in Appendix B to correlate surface-crack fracture data from the literature on a brittle epoxy material. In these data, the ratios of crack depth to plate thickness ranged from 0.15 to 1.0 and the ratios of crack depth to crack length ranged from 0.3 to 0.84. The equation correlated 95% of the data analyzed to within $\pm 10\%$ of the calculated failure stress.

The stress-intensity factor equations presented herein should be useful for correlating fatigue-crack-growth rates as well as in computing fracture toughness of surface-cracked plates.

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APPENDIX A

Fatigue-crack-growth patterns of surface cracks

The stress-intensity factor equation (eqn 1) developed for surface cracks is used herein to predict fatigue-crack-growth patterns under tension and bending fatigue loads. The predicted growth patterns are compared with experimental data obtained from the literature for steels, titanium alloys, and aluminum alloys.

Procedure. The surface-crack configuration considered is shown in Fig. 1. Although eqn (1) gives the stress-intensity factor at any location along the crack front, only the values at the maximum-depth point *A* and at the front surface *B* were used to predict the crack-growth patterns. (See insert in Fig. 1.) The cracks were always assumed to be semielliptical with semiaxes *a* and *c*.

The crack-growth rates were calculated by assuming that the Paris relationship[12] between crack-growth rate and stress-intensity factor range is obeyed independently at points *A* and *B* at the crack front. Thus,

$$\frac{da}{dN} = C_A \Delta K_A^n \quad (A1)$$

$$\frac{dc}{dN} = C_B \Delta K_B^n \quad (A2)$$

where ΔK is the stress-intensity factor range at point *A* or *B*, *n* is an exponent to be specified, and C_A and C_B are the crack-growth coefficients for points *A* and *B*, respectively. In this paper, *n* is assumed to be 4, a value which has been found to be applicable to a wide range of materials. Normally, C_A and C_B are assumed to be equal; however, experimental results[13, 14] for surface cracks under tension and bending fatigue loads show that small semicircular cracks tend to grow semicircular for low *a/t* ratios. Because the stress-intensity factor solution for the small semicircular crack shows that the stress intensity at point *B* is about 10% higher than the value at point *A*, the coefficient C_B was assumed to be

$$C_B = 0.9^n C_A \quad (A3)$$

so that a small semicircular crack would be predicted to initially retain its shape. Accordingly, eqn (A3) was used for all crack configurations considered. One reason C_A is not equal to C_B may be the changing relationship between the stress-intensity factor and the crack-growth rate as the stress state changes from plane stress on the front surface to plane strain at the maximum-depth point.

The number of stress cycles required for propagation of a surface crack from an initial half-length c_0 to a desired half-length c_f was obtained by a numerical integration of eqn (A2). This was accomplished by dividing the crack extension ($c_f - c_0$) into a large number of equal increments Δc and assuming that each increment was created at a constant crack-growth rate. The constant growth rate for each increment was determined from eqn (A2) by using the crack configuration which existed at the start of that growth increment. For each increment of crack advance Δc at the surface, a new increment of crack depth Δa was computed from

$$\Delta a = \frac{C_A}{C_B} \left(\frac{\Delta K_A}{\Delta K_B} \right)^n \Delta c = \left(\frac{\Delta K_A}{0.9 \Delta K_B} \right)^n \Delta c \quad (A4)$$

This defined the crack configuration for the next growth increment, and the process was repeated until the crack depth reached the plate thickness.

Tension. Figure 5 shows the experimental and predicted fatigue-crack-growth patterns for surface cracks subjected to tension. The figure shows the *a/c* ratio plotted vs the *a/t* ratio for Ti-6Al-4V titanium alloy[15], 9% nickel steel[16] and 2219-T87 aluminum alloy.

The experimental procedure for the aluminum and titanium alloys was as follows. An electric-discharge machined (EDM) notch was used as a crack starter. The vertical bar in Fig. 5 denotes the range of EDM notch shapes (*a/c*) for the nine aluminum-alloy specimens. Each specimen was subjected to constant-amplitude cyclic loading for various numbers of cycles and then statically pulled to failure. The data points indicate the final fatigue-crack shapes and sizes measured from the broken specimens. Hence, a separate specimen was necessary to obtain each data point.

The experimental data for the nickel steel were obtained from one specimen which was subjected to two-level variable-amplitude loading and then pulled to failure. The amplitude change caused "marker bands" to be formed on the crack plane. The marker bands, in turn, were used to define the crack shape and size.

The dashed curves in Fig. 5 are the predicted fatigue-crack-growth patterns from eqn (1) and eqns (A1)–(A4). (Note that *n* is assumed to be equal to 4 and that the growth patterns are independent of the magnitude of C_A and C_B . See eqn A4.)

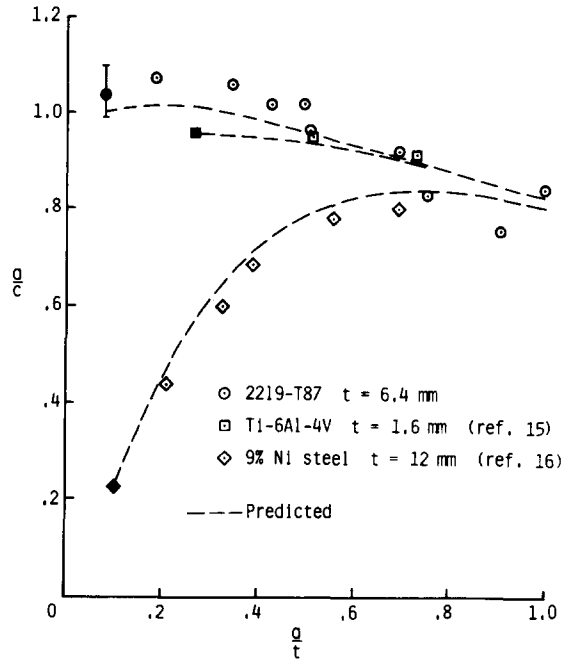


Fig. 5. Experimental and predicted fatigue-crack growth patterns for a surface crack in a plate under tension. (Solid symbols denote initial conditions.)

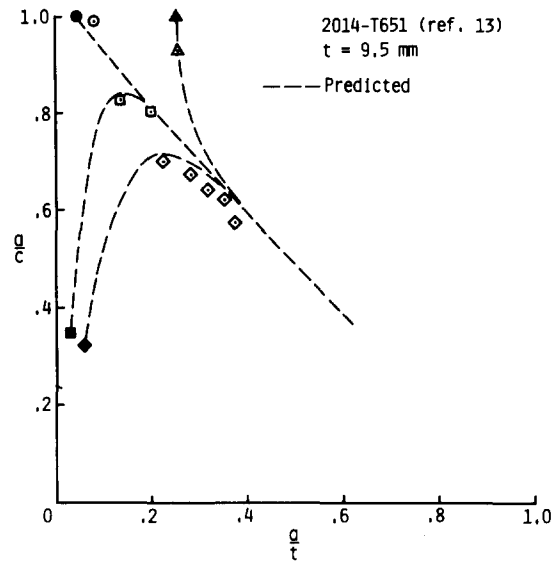


Fig. 6. Experimental and predicted fatigue-crack growth patterns for a surface crack in an aluminum alloy cantilever plate under bending. (Solid symbols denote initial conditions.)

The predicted growth patterns are in good agreement with the measurements made on the three materials. The solid symbols denote initial crack size and shape. For all initial crack shapes considered, the predicted a/c ratio was about 0.8 when the crack depth became equal to the plate thickness.

Bending. Figure 6 shows the experimental and predicted fatigue-crack-growth patterns for surface cracks in plates subjected to cantilever bending. Although the stress-intensity factor equation (eqn 1) used herein was developed for pure bending, the differences between crack-growth patterns for cantilever and pure bending are not expected to be large. The a/c ratio is plotted as a function of the a/t ratio for aluminum-alloy specimens. All data points for the 2014-T651 aluminum alloy [13] were obtained from separate specimens. The specimens were cycled under constant-amplitude loading and then statically pulled to failure. Again, the solid symbols denote the initial crack dimensions.

In Fig. 6, the predicted fatigue-crack-growth patterns (dashed curves) are in good agreement with the experimental data. The predicted results show that the cracks tended to approach a common propagation pattern, as pointed out by Corn [13].

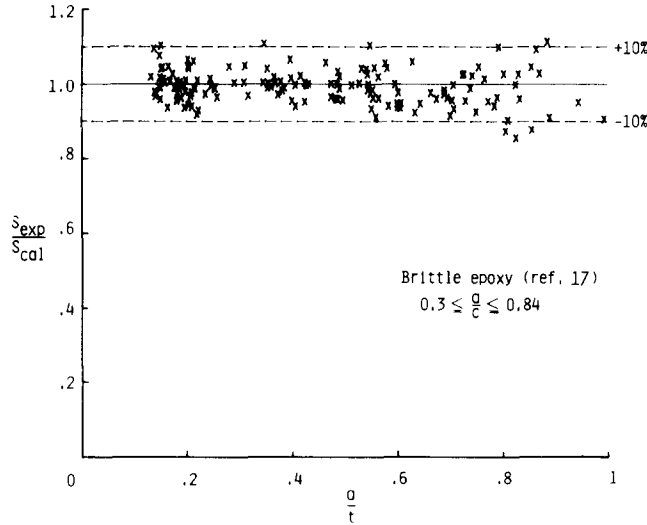


Fig. 7. Correlation of experimental failure stress S_{exp} to calculated failure stress S_{cal} for a brittle epoxy material as a function of a/t .

APPENDIX B

Fracture of surface-cracked brittle materials

The application of linear-elastic fracture mechanics to surface-cracked specimens is complicated because the stress-intensity factor solution is a function of the parametric angle ϕ . For surface cracks with $a/c < 0.6$, the maximum stress-intensity factor occurs at the maximum-depth point, $\phi = \pi/2$; for surface cracks with $a/c \geq 0.6$, the maximum stress-intensity factor occurs near the front surface $\phi = 0$. In fracture analyses, most investigators have used the stress-intensity factor at the maximum-depth point [9]. However, for a/c ratios greater than about 0.6, some investigators have used the value at the front surface because the stress-intensity factor is maximum there. In this paper, the maximum stress-intensity factors from eqn (1) were used. The maximum value occurred at either $\phi = 0$ or $\pi/2$.

Equation (1) was used to analyze data obtained by Smith [17] in a large number of fracture tests on surface-cracked tension specimens made of a brittle epoxy material. (Plane-strain plastic-zone size, based on the largest computed fracture toughness, was two orders of magnitude below minimum specimen thickness and, hence, consistent with brittle fracture behavior.) Thicknesses ranged from 2.5 to 9.5 mm, with $0.15 \leq a/t \leq 1.0$ and $0.3 \leq a/c \leq 0.84$. All specimens were 25-mm wide.

In this analysis, the specimens were arranged into five groups according to their date of manufacture. Specimen thicknesses were constant within each group. Fracture of all specimens was assumed to occur at the same value of stress-intensity factor (denoted herein by K_{cr}). The elastic fracture toughness K_{cr} for each group of specimens was obtained by averaging the calculated stress-intensity factor at failure K_{Ic} as

$$K_{cr} = \frac{1}{m} \sum_{i=1}^m (K_{Ic})_i \quad (\text{B1})$$

where m is the number of specimens in a group. The K_{cr} values for the five groups of specimens were calculated as 675, 676, 713, 720 and 725 kN/m^{3/2}.

After K_{cr} was determined, eqn (1) was used to calculate failure stresses. The gross failure stresses S_{cal} were calculated from

$$S_{cal} = \frac{K_{cr}}{\sqrt{\pi \frac{a}{Q} F_{max}}} \quad (\text{B2})$$

where F_{max} denotes the maximum value of F for a given a/c , a/t and c/b . Figure 7 shows the ratio of experimental failure stress S_{exp} to calculated failure stress S_{cal} plotted as a function of the a/t ratio. The solid line at unity denotes perfect agreement and the dashed lines denote $\pm 10\%$ scatter. The proposed equation correlated 95% of the data analyzed within $\pm 10\%$ for a wide range of a/t and a/c ratios.